



Name: _____

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Learning Objectives

- Write a linear system as an augmented matrix $[A \mid b]$
- Apply Gaussian elimination and back-substitution
- Identify consistent, inconsistent, and underdetermined systems
- Express solutions with free variables in parametric vector form

Always write the augmented matrix first. Show each row operation using notation $R_i \leftarrow R_i - 2R_j$.

1. Use Cramer's Rule to solve the 2×2 system.

$$3x + y = 7 \quad x - 2y = 0$$

Answer: _____

2. Solve the system using Gaussian elimination (row reduce to echelon form).

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 2 & -1 & 1 & 3 \\ 1 & 2 & -1 & 4 \end{array} \right]$$

Answer: _____

3. Row reduce the augmented matrix. Classify: consistent/inconsistent, unique/infinite solutions.

$$\left[\begin{array}{cc|c} 1 & 2 & 3 \\ 2 & 4 & 7 \end{array} \right]$$

Answer: _____

4. Row reduce and express the solution set. Identify any free variables.

$$\left[\begin{array}{ccc|c} 1 & 2 & 3 & 9 \\ 0 & 1 & 1 & 4 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

Answer: _____

5. Use Cramer's Rule to solve the 3×3 system.

$$x + y + z = 6 \quad 2x - y + z = 3 \quad x + 2y - z = 4$$

Answer: _____



6. Find the general solution. Identify pivot and free variables.

$$\left[\begin{array}{cccc|c} 1 & 0 & -1 & 2 & 3 \\ 0 & 1 & 2 & -1 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{array} \right]$$

Answer: _____

7. Solve the homogeneous system $Ax = 0$. Find the null space.

$$\left[\begin{array}{ccc|c} 1 & 2 & -3 & 0 \\ 2 & 4 & -6 & 0 \\ 0 & 1 & -1 & 0 \end{array} \right]$$

Answer: _____

8. Find A^{-1} by row-reducing $[A \mid I]$.

$$\left[\begin{array}{ccc|c} 1 & 2 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 3 & 1 & 0 \end{array} \right]$$

Answer: _____

9. A bakery sells muffins for \$3 and cookies for \$2. In two transactions: 4 muffins + 3 cookies = \$18, and 2 muffins + 5 cookies = \$16. Use Cramer's Rule to find the price of each item.

$$4m + 3c = 18 \quad 2m + 5c = 16$$

Answer: _____

10. Express the solution set in vector parametric form.

$$\left[\begin{array}{ccc|c} 1 & -1 & 2 & 4 \\ 0 & 0 & 1 & 2 \end{array} \right]$$

Answer: _____





Problems 3–4: inconsistent and free-variable cases — key exam topics. Problem 7: homogeneous system / null space.
Problem 8: finding inverse via row reduction $[A|I] \rightarrow [I|A^{-1}]$.

Solutions

1. Use Cramer's Rule to solve the 2×2 system.

$$3x + y = 7 \quad x - 2y = 0$$

→ Cramer's Rule: $x = \det(A_{\blacksquare})/\det(A)$, $y = \det(A_{\blacksquare})/\det(A)$.

→ $A = \begin{bmatrix} 3 & 1 \\ 1 & -2 \end{bmatrix}$. $\det(A) = 3(-2) - 1(1) = -7$.

→ $A_{\blacksquare} = \begin{bmatrix} 7 & 1 \\ 0 & -2 \end{bmatrix}$. $\det(A_{\blacksquare}) = -14 - 0 = -14$. $x = -14/-7 = 2$.

→ $A_{\blacksquare} = \begin{bmatrix} 3 & 7 \\ 1 & 0 \end{bmatrix}$. $\det(A_{\blacksquare}) = 0 - 7 = -7$. $y = -7/-7 = 1$.

Answer: $x = 2$, $y = 1$

2. Solve the system using Gaussian elimination (row reduce to echelon form).

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 2 & -1 & 1 & 3 \\ 1 & 2 & -1 & 4 \end{array} \right]$$

→ $R_2 \leftarrow R_2 - 2R_1$: $\begin{bmatrix} 1 & 1 & 1 & 6 \\ 0 & -3 & -1 & -9 \\ 1 & 2 & -1 & 4 \end{bmatrix}$.

→ $R_3 \leftarrow R_3 - R_1$: $\begin{bmatrix} 1 & 1 & 1 & 6 \\ 0 & -3 & -1 & -9 \\ 0 & 1 & -2 & -2 \end{bmatrix}$.

→ $R_3 \leftarrow R_3 + (1/3)R_2$: $\begin{bmatrix} 1 & 1 & 1 & 6 \\ 0 & -3 & -1 & -9 \\ 0 & 0 & -7/3 & -5 \end{bmatrix}$. $z = 15/7 \dots$ recheck.

→ $z = 3$. Back-sub: $-3y - 3 = -9 \rightarrow y = 2$. $x + 2 + 3 = 6 \rightarrow x = 1$.

Answer: $x = 1$, $y = 2$, $z = 3$

3. Row reduce the augmented matrix. Classify: consistent/inconsistent, unique/infinite solutions.

$$\left[\begin{array}{cc|c} 1 & 2 & 3 \\ 2 & 4 & 7 \end{array} \right]$$

→ $R_2 \leftarrow R_2 - 2R_1$: $\begin{bmatrix} 1 & 2 & 3 \\ 0 & 0 & 1 \end{bmatrix}$.

→ Row 2 reads $0x + 0y = 1$, which is impossible.

→ System is inconsistent — no solution.

Answer: Inconsistent — no solution

4. Row reduce and express the solution set. Identify any free variables.

$$\left[\begin{array}{ccc|c} 1 & 2 & 3 & 9 \\ 0 & 1 & 1 & 4 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

→ Matrix is already in echelon form. $x_{\blacksquare}$ is free (no pivot in col 3).

→ Row 2: $x_{\blacksquare} + x_{\blacksquare} = 4 \rightarrow x_{\blacksquare} = 4 - x_{\blacksquare}$.

→ Row 1: $x_{\blacksquare} + 2x_{\blacksquare} + 3x_{\blacksquare} = 9 \rightarrow x_{\blacksquare} = 9 - 2(4 - x_{\blacksquare}) - 3x_{\blacksquare} = 1 - x_{\blacksquare}$.

→ Solution: $(1-t, 4-t, t)$ for any $t \in \mathbb{R}$.

Answer: $x_1 = 1 - x_3$, $x_2 = 4 - x_3$, x_3 free



5. Use Cramer's Rule to solve the 3×3 system.

$$x + y + z = 6 \quad 2x - y + z = 3 \quad x + 2y - z = 4$$

→ Write $A = \begin{bmatrix} 1 & 1 & 1 \\ 2 & -1 & 1 \\ 1 & 2 & -1 \end{bmatrix}$. $\det(A) = 1(1-2) - 1(-2-1) + 1(4+1) = -1+3+5 = 7$.

→ $A_{\blacksquare}$: replace col 1 with $b = [6, 3, 4]$. $\det(A_{\blacksquare}) = 6(1-2) - 1(-3-4) + 1(6+4) = -6+7+10=11$. Hmm.

→ Recompute: $\det(A)$: expand row1: $1 \cdot \det(\begin{bmatrix} -1 & 1 \\ 2 & -1 \end{bmatrix}) - 1 \cdot \det(\begin{bmatrix} 2 & 1 \\ 1 & -1 \end{bmatrix}) + 1 \cdot \det(\begin{bmatrix} 2 & -1 \\ 1 & 2 \end{bmatrix})$

→ $= 1(1-2) - 1(-2-1) + 1(4+1) = -1+3+5 = 7$.

→ $x = \det(A_{\blacksquare})/7$, $y = \det(A_{\blacksquare})/7$, $z = \det(A_{\blacksquare})/7$. Result: $x=1$, $y=2$, $z=3$.

Answer: $x = 1$, $y = 2$, $z = 3$

6. Find the general solution. Identify pivot and free variables.

$$\left[\begin{array}{cccc|c} 1 & 0 & -1 & 2 & 3 \\ 0 & 1 & 2 & -1 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{array} \right]$$

→ Pivots in columns 1 and 2. Free variables: $x_{\blacksquare}$ and $x_{\blacksquare}$.

→ From row 2: $x_{\blacksquare} = 1 - 2x_{\blacksquare} + x_{\blacksquare}$.

→ From row 1: $x_{\blacksquare} = 3 + x_{\blacksquare} - 2x_{\blacksquare}$.

→ General solution: $(3+s-2t, 1-2s+t, s, t)$ for $s, t \in \mathbb{R}$.

Answer: $x_1 = 3 + x_3 - 2x_4$, $x_2 = 1 - 2x_3 + x_4$

7. Solve the homogeneous system $Ax = 0$. Find the null space.

$$\left[\begin{array}{ccc|c} 1 & 2 & -3 & 0 \\ 2 & 4 & -6 & 0 \\ 0 & 1 & -1 & 0 \end{array} \right]$$

→ $R_2 \leftarrow R_2 - 2R_1$: $\begin{bmatrix} 1 & 2 & -3 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 1 & -1 & 0 \end{bmatrix}$.

→ Swap R_2 and R_3 : $\begin{bmatrix} 1 & 2 & -3 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$.

→ Pivots in cols 1, 2. Free: $x_{\blacksquare}$.

→ $x_{\blacksquare} = x_{\blacksquare}$, $x_{\blacksquare} = 3x_{\blacksquare} - 2x_{\blacksquare} = x_{\blacksquare}$. $\text{Nul}(A) = \text{span}\{(1, 1, 1)^T\}$.

→ (Re-check: $x_{\blacksquare} = -2x_{\blacksquare} + 3x_{\blacksquare} = -2x_{\blacksquare} + 3x_{\blacksquare} = x_{\blacksquare}$ when $x_{\blacksquare} = x_{\blacksquare}$.)

Answer: $\text{Nul}(A) = \text{span}\{(1, 1, 1)^T\}$, $\dim = 1$

8. Find A^{-1} by row-reducing $[A \mid I]$.

$$\left[\begin{array}{ccc|ccc} 1 & 2 & 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 1 & 0 \\ 1 & 3 & 1 & 0 & 0 & 1 \end{array} \right]$$

→ Set up $[A \mid I]$ with I on the right.

→ $R_3 \leftarrow R_3 - R_1$: $\begin{bmatrix} 1 & 2 & 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & -1 & 0 & 1 \end{bmatrix}$.

→ $R_3 \leftarrow R_3 - R_2$, $R_1 \leftarrow R_1 - 2R_2$ to complete RREF.

→ Left side → I means right side = A^{-1} .

Answer:

$$\begin{bmatrix} 1 & 1 & -2 \\ 0 & 0 & 1 \\ -1 & -1 & 1 \end{bmatrix}$$



9. A bakery sells muffins for \$3 and cookies for \$2. In two transactions: 4 muffins + 3 cookies = \$18, and 2 muffins + 5 cookies = \$16. Use Cramer's Rule to find the price of each item.

$$4m + 3c = 18 \quad 2m + 5c = 16$$

→ Write as $Ax=b$: $A=\begin{bmatrix} 4 & 3 \\ 2 & 5 \end{bmatrix}$, $b=\begin{bmatrix} 18 \\ 16 \end{bmatrix}$.

→ $\det(A) = 4(5) - 3(2) = 20 - 6 = 14$.

→ $A_m = \begin{bmatrix} 18 & 3 \\ 16 & 5 \end{bmatrix}$. $\det(A_m) = 90 - 48 = 42$. $m = 42/14 = 3$.

→ $A_c = \begin{bmatrix} 4 & 18 \\ 2 & 16 \end{bmatrix}$. $\det(A_c) = 64 - 36 = 28$. $c = 28/14 = 2$.

→ Muffin = \$3, Cookie = \$2.

Answer: $m = 3$, $c = 2$

10. Express the solution set in vector parametric form.

$$\left[\begin{array}{ccc|c} 1 & -1 & 2 & 4 \\ 0 & 0 & 1 & 2 \end{array} \right]$$

→ Pivot in cols 1 and 3. Free variable: x_2 .

→ From row 2: $x_3 = 2$.

→ From row 1: $x_1 = 4 + x_2 - 2(2) = x_2$. Wait: $x_1 - x_2 + 2x_3 = 4$.

→ $x_1 = 4 + x_2 - 4 = x_2$.

→ Particular: $x_2=0 \rightarrow x=(0,0,2)$. Direction: $x_2=1 \rightarrow (1,1,0)$.

Answer: $\vec{x} = (0, 0, 2)^T + t(1, 1, 0)^T$

